

I. Give more strings in L aside from  $w = 01010101$ . Reg Exp for  $L = (01)^*(10)^*0(10)^*1(01)^*$

1.  $1010101010$  matches  $(10)^*$

2.  $0101010$  matches  $0(10)^*$

3.  $101010101$  matches  $1(01)^*$

4. "" matches  $(01)^*(10)^*0(10)^*1(01)^*$

5.  $0101$  matches  $(01)^*$

6.  $1010$  matches  $(10)^*$

7.  $010$  matches  $0(10)^*$

8.  $101$  matches  $1(01)^*$

9.  $1$  matches  $1(01)^*$

10.  $0$  matches  $0(10)^*$

II. **RE =  $1^*00^*1(0+1)^*$**

Task:

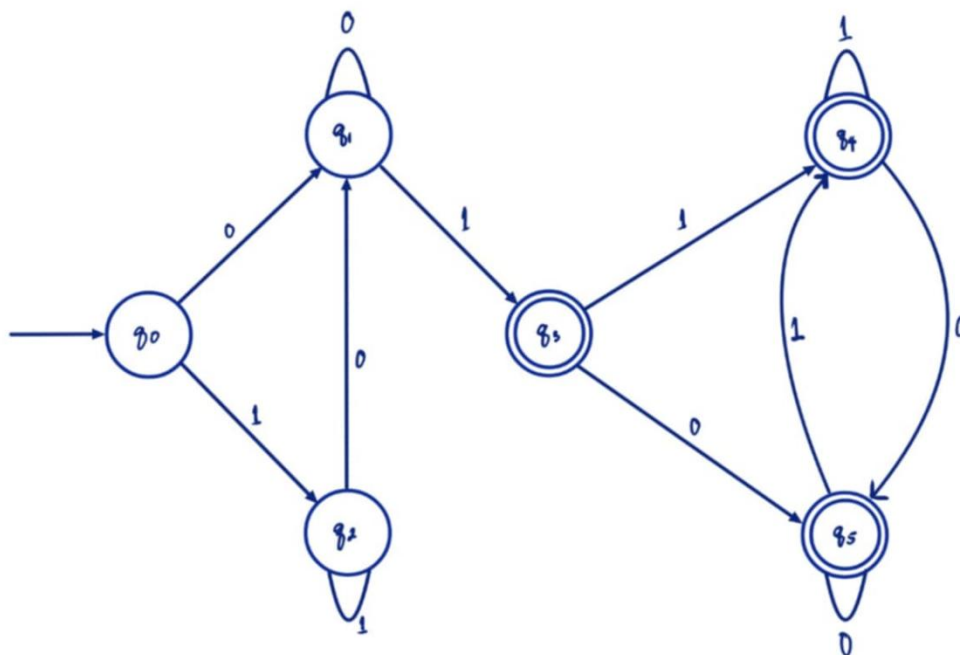
- i. Construct one full valid string that matches the regular expression by explicitly applying each operator ( $\{^*\}$ ,  $\{+\}$ ).
  1. Indicate clearly which parts of the regular expression.

Answer: **11100111**

where:

- 111** matches  $1^*$  in  $1^*00^*1(0+1)^*$   
**0** matches 0 in  $1^*00^*1(0+1)^*$   
**0** matches  $0^*$  in  $1^*00^*1(0+1)^*$   
**1** matches 1 in  $1^*00^*1(0+1)^*$   
**11** matches  $1^*$  in  $1^*00^*1(0+1)^*$

- ii. Draw the DFA that accepts the same string:



1. Label all states and transitions,
2. Identify the start and accepting state(s),

| State/Input      | 0               | 1               |
|------------------|-----------------|-----------------|
| ->q <sub>0</sub> | q <sub>1</sub>  | q <sub>2</sub>  |
| q <sub>1</sub>   | q <sub>1</sub>  | *q <sub>3</sub> |
| q <sub>2</sub>   | q <sub>1</sub>  | q <sub>2</sub>  |
| *q <sub>3</sub>  | *q <sub>5</sub> | *q <sub>4</sub> |
| *q <sub>4</sub>  | *q <sub>5</sub> | *q <sub>4</sub> |
| *q <sub>5</sub>  | *q <sub>5</sub> | *q <sub>4</sub> |

q<sub>0</sub> is the starting state.

\*q<sub>3</sub>, \*q<sub>4</sub>, and \*q<sub>5</sub> are the accepting states.

3. Make sure the DFA accepts the constructed string in Question #1

*Transition Equation Guide:*

$\delta(\text{current state, input symbol}) = \text{next state}$

String: **11100111**

$$\delta(q_0, 1) = q_2$$

$$\delta(q_2, 1) = q_2$$

$$\delta(q_2, 1) = q_2$$

$$\delta(q_0, 0) = q_1$$

$$\delta(q_0, 0) = q_1$$

$$\delta(q_0, 1) = q_3$$

$$\delta(q_5, 1) = q_5$$

$$\delta(q_5, 1) = q_5$$

The string **11100111** is a valid string and is accepted by the DFA.

### III. Machine Code Translation Task

Based on the sample CFG production rules presented in the project, consider the context-free language that generates the following variable declaration and assignment:

x: int  
x = 100

Task: Write the equivalent machine-level code for this statement using an assembly language of your choice (e.g., x86, MIPS, or ARM), based on your understanding of instruction-level representation from your *Computer Architecture and Organization* course.

```
1  ; ALA-AN, PRINCESS BEVERLY
2
3  section .data          ; defining and reserve space for variables in memory
4      x dd 0             ; declaring x
5      ; [label] mnemonic [operands]
6      ; dd means 'define doubleword' which allocates 4 bytes
7      ; I initially store 0 value in x
8      ; x: int
9
10 section .text          ; actual instructions starts here
11     global _start      ; execution begins
12
13 _start:
14     mov dword [x], 100  ; assigning 100 to x
15     | | | |           ; x = 100
16
17
18     mov eax, 1          ; eax = 1 means system call to exit the program
19     xor ebx, ebx        ; exit code
20     int 0x80            ; ending the program successfully
```